

Under review as a conference paper at ICAIS 2025

CAUSAL-INFORMED ADAPTIVE LEARNING FOR
CONTEXTUAL PERSONALIZATION IN
RECOMMENDATION SYSTEMS

Anonymous authors
Paper under double-blind review

ABSTRACT

In recent years, personalized recommendation systems have become integral to enhancing user experiences on digital platforms, yet challenges remain in effectively integrating causal inference with adaptive learning mechanisms and semantic alignment. Traditional systems predominantly rely on correlation-based models, often overlooking the dynamic causal relationships within user interaction data that could enhance recommendation precision and contextual relevance. This paper addresses these gaps by presenting a novel framework that synergizes causal inference using structural equation models and causal diagrams, adaptive learning algorithms via a refined hybrid multi-armed bandit strategy, and semantic content mapping with advanced natural language processing techniques such as Latent Dirichlet Allocation and BERT-based embeddings. Through this integrated approach, our method dynamically adjusts recommendations to align with user preferences and adapt to context changes. Empirical evaluation demonstrates our method’s superiority in achieving higher accuracy and relevance in personalized content delivery compared to existing models. The findings underscore the potential of our framework to significantly improve recommendation cohesion and user satisfaction, marking a substantial advancement in the field of contextual personalization.

1 INTRODUCTION

1.1 RESEARCH BACKGROUND

In recent years, recommendation systems have undergone significant advancements, driven by the imperative to personalize user experiences across digital platforms such as e-commerce, streaming services, and content delivery networks. A substantial portion of contemporary research in this field focuses on utilizing user interaction data to bolster user engagement through personalized content delivery. Causal inference plays an increasingly crucial role in elucidating the underlying cause-effect dynamics within this data, offering deeper insights beyond mere correlations (Pearl, 2009). Traditional recommendation systems have predominantly relied on correlation-based machine learning models. These models are now being supplemented with structural equation modeling (SEM) and causal diagrams to improve user profiling and interpretability (Shmueli, 2010).

Adaptive learning algorithms have also garnered considerable attention, particularly those addressing the exploration-exploitation trade-off in recommendation systems. Multi-armed bandit (MAB) frameworks, complemented by strategies such as Thompson Sampling and epsilon-greedy models, have demonstrated efficacy in dynamically refining recommendations based on user feedback (Gittins et al., 2011). Additionally, semantic content mapping has become important in ensuring contextual alignment with user preferences. Techniques from natural language processing (NLP), such as Latent Dirichlet Allocation (LDA) and contextual embeddings, are employed to achieve this



Meta Review Status: The meta review and decision have not been released yet.

AI Review from DeepReviewer



AI Review from DeepReviewer will be automatically processed



Summary

This paper introduces a novel framework for personalized recommendation systems, aiming to enhance the accuracy and relevance of recommendations by integrating causal inference, adaptive learning, and semantic content mapping. The core idea is to move beyond traditional correlation-based approaches by incorporating causal relationships, adapting to user feedback through a multi-armed bandit (MAB) strategy, and aligning recommendations with user preferences using semantic analysis of content. The proposed methodology involves several key steps. First, causal inference is employed to identify significant causal variables from user interaction data, using structural equation models (SEMs) and causal diagrams. These causal variables are then used to inform the adaptive learning component, which utilizes a hybrid MAB strategy to balance exploration and exploitation. Finally, semantic content mapping is achieved through natural language processing (NLP) techniques, specifically Latent Dirichlet Allocation (LDA) for topic modeling and BERT-based contextual embeddings to capture nuanced user contexts. The paper presents an experimental evaluation comparing the proposed method against several baselines, demonstrating improvements in accuracy, precision, recall, and F1-score. The authors claim that their integrated approach provides a more holistic and effective way to personalize recommendations, addressing the limitations of traditional methods. While the paper presents a promising approach, my analysis reveals several critical areas that require further attention, particularly regarding scalability, implementation details, and the rigor of the experimental evaluation. The lack of specific details on the implementation of the NLP techniques and the computational complexity of the framework, combined with the limited information about the experimental setup, raises concerns about the practical applicability and reproducibility of the proposed method. Despite these limitations, the paper's attempt to integrate causal inference, adaptive learning, and semantic content mapping represents a significant step towards more sophisticated and context-aware recommendation systems.



Strengths

The primary strength of this paper lies in its conceptual framework, which integrates causal inference, adaptive learning, and semantic content mapping into a unified approach for personalized recommendations. This is a significant contribution, as it moves beyond traditional correlation-based models by explicitly considering causal relationships, adapting to user feedback, and aligning recommendations with user preferences at a semantic level. The use of structural equation models

(SEMs) and causal diagrams to identify significant causal variables is a notable strength, as it allows for a deeper understanding of the underlying causal relationships within user interaction data. This approach has the potential to improve user profiling and recommendation strategies by moving beyond simple correlations. Furthermore, the development of a hybrid multi-armed bandit (MAB) strategy that incorporates causal insights is another positive aspect. This strategy aims to balance exploration and exploitation more effectively, potentially leading to improved recommendation relevance and precision. The paper also advances semantic content mapping by utilizing natural language processing (NLP) techniques such as Latent Dirichlet Allocation (LDA) and BERT-based contextual embeddings. This ensures that recommendations are contextually aligned with user preferences, fostering enhanced user engagement. The combination of these three components—causal inference, adaptive learning, and semantic content mapping—represents a holistic approach to personalized recommendation systems, addressing the limitations of traditional methods. The paper's attempt to integrate these diverse techniques into a cohesive framework is a novel contribution to the field. The experimental results, while limited in detail, do show improvements in accuracy, precision, recall, and F1-score compared to the baselines, providing some empirical support for the proposed approach. The idea of using causal insights to inform the adaptive learning process is particularly innovative and could lead to more robust and effective recommendation systems. Overall, the paper's strength lies in its conceptual framework and the potential of its integrated approach to advance the field of personalized recommendations.

Weaknesses

My analysis reveals several critical weaknesses in this paper, primarily concerning the lack of detail regarding implementation, scalability, and the rigor of the experimental evaluation. First, the paper lacks a detailed discussion on the scalability of the proposed framework. While the paper describes the use of structural equation models (SEMs), multi-armed bandit (MAB) algorithms, and BERT-based embeddings, it fails to address the computational complexity of these components, especially when dealing with large datasets or real-time recommendation scenarios. The paper does not specify the time complexity of the SEM estimation or the MAB updates, making it difficult to assess the practical feasibility of the approach for large-scale applications. For instance, the paper mentions "Post-training, the model outputs key causal elements vital for enhancing recommendation strategies and precision" (Section 3.1), but does not detail the computational cost of this process. Similarly, the "Technical Integration" subsection (Section 3.2) states that "Causal features identified via SEMs and diagrams are integrated, ensuring recommendations align with user preferences and contexts," but lacks details on the computational implications of this integration. The absence of a discussion on resource requirements and potential bottlenecks, such as the training of the BERT model, the estimation of the causal models, and the online adaptation of the MAB strategy, is a significant oversight. This lack of attention to scalability raises serious concerns about the practical applicability of the proposed framework in real-world settings. My confidence in this weakness is high, as the paper completely omits any discussion of computational complexity or resource requirements. Second, the paper does not provide sufficient details on the implementation of the NLP techniques used for semantic content mapping. While the paper mentions the use of Latent Dirichlet Allocation (LDA) and BERT-based contextual embeddings (Section 3.3), it fails to specify the exact BERT variant used (e.g., base, large, or a specific fine-tuned version) and the parameters used for LDA (e.g., number of topics, alpha, and beta values). The paper also lacks a discussion on the pre-processing steps applied to the text data before feeding it into the NLP models, which is crucial for reproducibility and understanding the overall pipeline. For example, the "OPERATIONAL PROCESS" subsection (Section 4.3.2) mentions "tokenization and topic space projection," but does not detail other pre-processing steps like stemming or stop-word removal. Furthermore, the paper does not clarify how the semantic embeddings are used in the recommendation process, whether they are used as features in the causal model or directly in the multi-armed bandit algorithm. The paper states that "Semantic content mapping is achieved through

advanced NLP techniques like Latent Dirichlet Allocation (LDA) and BERT-based contextual embeddings, ensuring that recommendations remain relevant and contextually aligned with evolving user preferences" (Section 1.3), but the mechanism of integration is not detailed. This lack of clarity makes it difficult to understand how the semantic information is used to influence the recommendation process. My confidence in this weakness is high, as the paper omits crucial implementation details that are essential for reproducibility and understanding the framework's inner workings. Finally, the experimental evaluation could be more comprehensive. The paper mentions using a dataset but does not specify its size, characteristics, or how it was preprocessed. The baselines are listed as "Traditional correlation-based models, standard multi-armed bandit frameworks without causal insights, and semantic content mapping techniques used independently," but the specific algorithms or implementations used for these baselines are not provided. The metrics are initially listed as "[Metric1, Metric2]" without specifying what these metrics are, although the example results show "Accuracy, Precision, Recall, F1-Score." The paper also mentions "Ablation studies highlighted each component's impact," suggesting they were performed, but the details of these studies (which components were ablated and the results) are not presented in the main text. This lack of detail makes it difficult to assess the validity of the results and the contribution of each component of the framework. My confidence in this weakness is medium, as the paper does provide some information about the experimental setup, but lacks the necessary details to fully evaluate the results. The absence of specific details about the dataset, baseline implementations, and ablation studies significantly weakens the experimental evaluation.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, to tackle the scalability concerns, the authors should provide a detailed analysis of the computational complexity of each component of their framework, including the BERT model training, the causal model estimation, and the multi-armed bandit updates. They should also discuss the memory requirements and the potential bottlenecks in the pipeline. This analysis should include the time complexity of the SEM estimation, the MAB updates, and the BERT embedding generation. Furthermore, it would be beneficial to include experiments on larger datasets to demonstrate the scalability of the approach. The authors could also explore techniques for optimizing the computational efficiency of their method, such as using approximate inference methods for the causal models or employing more efficient multi-armed bandit algorithms. A discussion on the trade-offs between accuracy and computational cost would also be valuable. For example, they could investigate the use of lighter versions of BERT or explore alternative methods for semantic mapping that are less computationally expensive. This would provide a more realistic assessment of the framework's practical feasibility. Second, regarding the NLP techniques, the authors should provide a more detailed description of the implementation, including the specific BERT variant used (e.g., base, large, or a specific fine-tuned version), the parameters used for LDA (e.g., number of topics, alpha, and beta values), and the pre-processing steps applied to the text data (e.g., tokenization, stemming, and stop-word removal). They should also discuss the rationale behind their choices and provide an analysis of the impact of these choices on the overall performance. For example, they could include experiments comparing different BERT variants or different LDA parameter settings. Furthermore, the authors should clarify how the semantic embeddings are integrated into the recommendation process, such as whether they are used as features in the causal model or as part of the user/item representations in the multi-armed bandit algorithm. A clear explanation of how the semantic information is used to influence the recommendation process is crucial for understanding the framework's effectiveness. This would enhance the reproducibility and transparency of the research. Finally, to improve the experimental evaluation, the authors should provide more details about the dataset used, including its size, characteristics, and the distribution of user interactions. They should also provide a clear description of the baseline models used for comparison, including their implementation details and hyperparameter settings. The authors should justify their choice of evaluation metrics and provide a more detailed analysis of the results,

including the statistical significance of the improvements over the baselines. Furthermore, they should conduct ablation studies to demonstrate the contribution of each component of the framework, such as the causal inference module, the adaptive learning module, and the semantic mapping module. This could involve removing each component individually and evaluating the performance of the remaining framework. The authors should also consider including a comparison with other state-of-the-art recommendation methods to better contextualize the performance of their approach. This would provide a more comprehensive understanding of the framework's strengths and weaknesses. These changes would significantly strengthen the paper and make it more impactful.

? Questions

Based on my analysis, I have several questions that I believe are critical for understanding the proposed framework. First, could the authors provide more details on the dataset used for the experiments? Information about the size, characteristics, and preprocessing steps would be helpful to assess the generalizability of the results. Specifically, knowing the number of users, items, and interactions, as well as the distribution of these interactions, would be valuable. Second, how does the proposed framework perform in terms of computational efficiency and scalability? Are there any limitations in deploying this method in real-time recommendation scenarios? I am particularly interested in understanding the time complexity of the SEM estimation, the MAB updates, and the BERT embedding generation, as well as the memory requirements of the framework. Third, can the authors elaborate on the integration of the three components (causal inference, adaptive learning, and semantic mapping)? How do these components interact, and what are the challenges in ensuring their seamless cooperation? I am especially interested in understanding how the causal features are used to inform the adaptive learning process and how the semantic embeddings are incorporated into the recommendation process. Finally, what are the potential limitations of the proposed method, and how do you plan to address them in future work? I am particularly interested in understanding the limitations of the causal inference component, the challenges of using MABs in complex recommendation scenarios, and the potential biases that might be introduced by the semantic mapping techniques. Addressing these questions would provide a more complete understanding of the proposed framework and its potential impact on the field of personalized recommendation systems.

Scores

Soundness: 1.75

Presentation: 1.5

Contribution: 1.75

Rating: 4.0

 Summary

The paper proposes a unified framework for recommendation systems that integrates (i) causal inference (via structural equation models and causal diagrams) to identify causal factors for user profiling, (ii) adaptive learning with a hybrid multi-armed bandit (MAB) combining Thompson Sampling and epsilon-greedy to balance exploration and exploitation while incorporating causal features, and (iii) semantic content mapping using LDA and BERT-based embeddings for contextual alignment. The authors claim improved accuracy and relevance over baselines and report ablations and error analysis.

 Strengths

- Addresses an important problem: causally informed, context-aware personalization that adapts online.
- Clear high-level motivation for integrating causal reasoning with bandits and semantic representations (Sections 1.2–1.4).
- Acknowledges the need for semantic coherence via LDA and BERT embeddings (Sections 3.3, 4.3), which is practically relevant for content-based scenarios.
- Conceptual positioning of causal insights guiding exploration–exploitation could inspire future work if made concrete (Section 1.4, Contribution 3).

 Weaknesses

- Severe mismatch between causal claims and implementation details: despite repeated assertions of using SEMs and causal diagrams (Sections 1.4, 3.1, 4.1.1–4.1.2), Algorithm 1 (Section 4.1.3) is a trivial `nn.Linear` model with multiple syntactic errors (e.g., `"import torch(nn as nn"`, `"optimizer = torch.train农业大学(self.params(), lr=lr)"`, `"loss_backward()"`), and no mechanism for SEM estimation, causal graph specification, identification, or adjustment.
- Experimental section is non-reproducible and unverifiable: placeholders are used for dataset, metrics, and statistical tests (Section 4.4: `"[Dataset Name]"`, `"[Metric1, Metric2]"`, `"[Test Method]"`, `"[Parameter Details]"`). The reported table lacks

context and comparators, rendering claims of superiority unsupported.

- Integration between components is unspecified: there is no clear description of how causal features are derived (e.g., from a DAG) and fed into the bandit, nor how semantic embeddings (LDA/BERT) become context in a contextual bandit or affect reward modeling (Sections 3.2–3.3, 4.2–4.3).
- Bandit methodology is underspecified and partly incorrect/ambiguous: Equation (1) appears to be the Beta posterior mean for Bernoulli rewards, but spacing and notation are garbled ("s u c c e s s e s"), and there is no derivation or treatment of contextual information (Sections 3.2, 4.2). A hybrid of Thompson Sampling and epsilon-greedy is proposed, but no algorithmic pseudocode, schedules, or update rules are given.
- Clarity and presentation issues: duplicated high-level descriptions across Sections 3 and 4; broken pseudocode; lack of concrete examples; no figures or DAGs showing the causal graph; no detailed related work on causal bandits or deconfounded recommenders to situate contributions.
- Missing critical methodological details: assumptions (e.g., back-door/front-door), confounder handling, exposure bias correction, offline counterfactual evaluation (IPS/DR), hyperparameters, seeds, hardware, and training budgets.
- No ablation or error analysis details despite being claimed (Section 4.4).

? Questions

- Please provide a concrete causal diagram (DAG) of your problem setting, including variables for user attributes, item/content features, context, action/exposure, and outcomes. Specify identification assumptions (e.g., back-door set, instrumental variables, or front-door criterion) and how you verify them.
- How are SEMs estimated in practice? What is the exact model class (linear, latent variable SEM, etc.), estimator (e.g., ML, 2SLS, DML), and objective? What data and instruments/confounders are used? How do you transition from SEM parameters to features consumed by the bandit?
- Is your bandit contextual? If so, what is the context vector at decision time, and how does it incorporate causal features and semantic embeddings? Please provide full algorithmic pseudocode (action selection, posterior updates, and the hybrid TS/epsilon-greedy schedule).
- What are the precise datasets (e.g., MovieLens 1M/20M, MIND, Amazon, Adressa)? Provide splits, logging/exposure policy characteristics, and whether the data are logged bandit feedback.
- How do you handle exposure and selection bias in offline evaluation? Do you use IPS, self-normalized IPS, doubly robust estimators, or counterfactual risk minimization? What propensity estimates are used?
- Specify all metrics (e.g., CTR, NDCG@K, Recall@K, precision@K, cumulative regret), and the statistical tests (e.g., paired t-test, bootstrap) with confidence intervals.
- Detail the LDA and BERT components: pretrained model names, tokenization, pooling strategies, fine-tuning regimen, embedding dimensions, and how these embeddings are integrated into the bandit state and/or reward model.
- Provide ablation studies isolating the effect of causal components, bandit choice, and semantic mapping, with quantitative tables and variance across seeds.
- Compare against strong baselines: LinUCB/LinTS, deep contextual bandits, IPS-weighted/deconfounded recommenders, and causal bandit methods. How does your method perform relative to these?

- Release corrected code for Algorithm 1 and the full system. The current snippet contains multiple errors and does not reflect SEM/causal diagram usage.

Limitations

- Reliance on correct causal assumptions and graph specification; misspecification can harm personalization and fairness.
- Scalability and latency of maintaining causal estimates and semantic embeddings in real-time bandit updates are non-trivial and unaddressed.
- Cold-start and non-stationarity: the framework’s robustness to rapidly shifting preferences or new items/users is not analyzed.
- Potential reinforcement of filter bubbles and overlooked fairness/disparity impacts if causal variables correlate with sensitive attributes.
- Offline counterfactual evaluation challenges: high variance IPS/DR estimators and the need for accurate propensities are not discussed.

Image Evaluation

Cross-Modal Consistency: 18/50

Textual Logical Soundness: 12/30

Visual Aesthetics & Clarity: 10/20

Overall Score: 40/100

Detailed Evaluation (≤ 500 words):

1. Cross-Modal Consistency

- Visual ground truth: Table 1 (Method, Accuracy, Precision, Recall, F1-Score). Trend: Proposed > Baseline on all metrics.
- Major 1: Missing visuals promised by text (no causal diagrams/SEMs shown). Evidence: “using structural equation models and causal diagrams” (Sec 3.1/4.1) but no figures provided.
- Major 2: Results section contains placeholders, blocking verification. Evidence: Sec 4.4: “Dataset Name”, “[Metric1, Metric2]”, “[Parameter Details]”, “[Test Method]”.

Major 3: Claims ablation and statistical tests without showing them. Evidence: Sec 4.4: "Ablation studies highlighted... Statistical analysis using [Test Method]..."

- Major 4: Bandit action-selection formula inconsistent with Thompson Sampling. Evidence: Sec 3.2: " $a^* = \arg \max \text{BetaSample}(Q(a), n_{\text{counts}}(a))$ ".
- Minor 1: Equation (1) has corrupted symbols/spacing that impedes reading. Evidence: " $\sum \text{s u c c e s s e s}$ ", " $\sum \text{t r i a l s}$ ".
- Minor 2: Table 1 caption lacks dataset/task context. Evidence: "Performance Evaluation on Dataset Name".

2. Text Logic

- Major 1: Flagship superiority claim unsupported by concrete experimental setup. Evidence: Abstract/Sec 5: "Empirical evaluation demonstrates our method's superiority..." vs Sec 4.4 placeholders.
- Major 2: Method duplication and redundancy (Sec 3 vs Sec 4) blurs the contribution. Evidence: Sections 3 and 4 present near-identical overviews.
- Major 3: "SEMs and causal diagrams" use is not specified (no identification strategy, no graph, no estimand). Evidence: Sec 3.1/4.1 only describe a linear layer.
- Minor 1: Mixing SEM language with a simple linear nn model is misleading. Evidence: Algorithm 1 is a single `nn.Linear`.
- Minor 2: No description of how causal features feed the bandit (feature mapping, reward model). Evidence: Sec 4.2.2 generic "inform adaptive learning algorithms".

3. Figure Quality

- Major 1: N/A (no figures). No Major issues found.
- Minor 1: Table 1 is too generic; lacks dataset, baselines, CIs, and units; "best" not marked as claimed. Evidence: Table 1 content.
- Minor 2: Algorithm 1 code block is error-ridden, hurting reproducibility readability. Evidence: "import torch(nn as nn", "torch.train农业大学", "loss_backward()".

Key strengths:

- Ambitious integration of causality, bandits, and semantic models.
- Clear high-level motivation and decomposition (causal → adaptive → semantic).

Key weaknesses:

- Results are unverifiable (placeholders, missing datasets/tests/ablations).
- Causal methodology unspecified (no DAG, estimand, identification/adjustment).
- Bandit math and code contain errors; algorithmic details insufficient.
- Redundant sections and vague integration between components.

Recommendations:

- Add concrete datasets, baselines, and full results with CIs and ablations.
- Include causal graph, identification strategy (e.g., back-door set), and estimators.
- Correct bandit equations (sample $\text{Beta}(\alpha a, \beta a)$, not from $Q(a)$); define reward model and context integration.

- Replace Algorithm 1 with valid code or precise pseudo-code; remove placeholders.

Scores

Originality: **2**

Quality: **1**

Clarity: **1**

Significance: **1**

Soundness: **1**

Presentation: **1**

Contribution: **1**

Rating: 3

AI Review from SafeReviewer

 AI Review from SafeReviewer will be automatically processed

Summary

This paper introduces a novel framework for personalized recommendation systems, aiming to integrate causal inference, adaptive learning, and semantic content mapping. The core idea is to leverage causal relationships derived from user interaction data to enhance the accuracy of user profiling, which then informs an adaptive learning algorithm based on a multi-armed bandit (MAB) approach. Specifically, the authors propose using Structural Equation Models (SEMs) and causal diagrams to identify causal variables that influence user preferences. These causal insights are then incorporated into a hybrid MAB strategy, combining Thompson Sampling and epsilon-greedy methods, to dynamically adjust recommendations. Furthermore, the framework employs Latent Dirichlet Allocation (LDA) for topic modeling and BERT-based embeddings to achieve semantic alignment between user preferences and recommended items. The authors claim that this integrated approach improves recommendation cohesion and user satisfaction by moving beyond simple correlation-based models to capture underlying causal relationships. The experimental evaluation, conducted on a single dataset, demonstrates the superiority of the proposed method over baseline models in terms of accuracy, precision, recall, and F1-score. The authors also perform ablation studies to highlight the contribution of each component. While the paper presents a promising approach, it suffers from significant issues in writing quality, including grammatical errors and unclear phrasing, which

hinder the clarity and readability of the work. Additionally, the paper lacks sufficient detail in the description of the proposed framework, particularly in the integration of the different components, and the experimental section lacks crucial details, such as the specific dataset used and the statistical tests employed. Despite these limitations, the paper's core idea of integrating causal inference with adaptive learning and semantic mapping is a valuable contribution to the field of personalized recommendation systems.

✓ Strengths

The primary strength of this paper lies in its conceptual framework, which integrates causal inference, adaptive learning, and semantic content mapping into a unified approach for personalized recommendation systems. This integration is a novel contribution, as it attempts to move beyond traditional correlation-based models by incorporating causal relationships derived from user interaction data. The use of Structural Equation Models (SEMs) and causal diagrams to identify causal variables influencing user preferences is a sound approach, as it allows for a deeper understanding of the underlying mechanisms driving user behavior. Furthermore, the combination of Thompson Sampling and epsilon-greedy strategies within a multi-armed bandit (MAB) framework provides a robust method for balancing exploration and exploitation in dynamic environments. The inclusion of semantic content mapping using Latent Dirichlet Allocation (LDA) and BERT-based embeddings is also a valuable addition, as it ensures that recommendations are contextually aligned with user preferences. The experimental results, while limited to a single dataset, demonstrate the potential of the proposed method, showing improvements in accuracy, precision, recall, and F1-score compared to baseline models. The ablation studies, although not detailed, also suggest that each component of the framework contributes to the overall performance. In summary, the paper's strength lies in its innovative combination of established techniques to address the complex problem of personalized recommendation, offering a promising direction for future research.

✗ Weaknesses

After a thorough review of the paper, I have identified several significant weaknesses that impact its overall quality and credibility. Firstly, the paper suffers from severe writing quality issues. There are numerous grammatical errors, awkward phrasing, and instances of unclear language throughout the text. For example, the abstract contains the sentence: "In recent years, personalized recommendation systems have become integral to enhancing user experiences on digital platforms,yet challenges remain in effectively integrating causal inference with adaptive learning mechanisms and semantic alignment." The comma after 'platforms' is unnecessary and should be removed. Similarly, the introduction contains sentences like: "In recent years,recommendation systems have undergone significant advancements, driven by the imperative to personalize user experiences across digital platforms such as e-commerce, streaming services,and content delivery networks." The comma after 'years' should be replaced with a period or semicolon, and the word 'recommender' is more appropriate than 'recommendation' when referring to the system itself. These examples highlight a lack of attention to detail in the writing,

which makes the paper difficult to read and understand. This is not just a minor issue; it significantly hinders the paper's clarity and professionalism. My confidence in this assessment is high, as the errors are readily apparent throughout the text.

Secondly, the paper lacks sufficient detail in the description of the proposed framework. While the authors outline the three main components—causal inference, adaptive learning, and semantic mapping—the precise mechanisms for their integration are not clearly explained. For instance, in section 3.2, the paper states: "Technical Integration: Causal features identified via SEMs and diagrams are integrated, ensuring recommendations align with user preferences and contexts. This sophisticated integration enhances the adaptive algorithm's precision and contextual relevance." However, the paper does not specify how these causal features are mathematically incorporated into the MAB framework. The paper also does not provide sufficient details on the implementation of the causal inference component. Section 4.1.3 provides a code snippet for a linear regression model, but it does not explain how Structural Equation Models (SEMs) and causal diagrams are implemented. The paper mentions using SEMs and causal diagrams to identify causal features, but it does not detail the specific techniques or algorithms used for this purpose. This lack of detail makes it difficult to understand the proposed method fully and to replicate the results. My confidence in this assessment is high, as the paper consistently lacks specific details in its descriptions of key components and their integration.

Thirdly, the experimental section is inadequate. The paper fails to specify the dataset used for evaluation, which is crucial for reproducibility and understanding the context of the results. Section 4.4 states: "To evaluate our framework's efficacy, we conducted experiments on [Dataset Name], utilizing metrics such as [Metric1, Metric2]." The use of placeholders like '[Dataset Name]' and '[Metric1, Metric2]' indicates that the experimental details are incomplete. Furthermore, the paper does not provide sufficient information about the baseline models used for comparison. While the paper mentions a 'Baseline method,' it does not specify the exact algorithms or configurations used for this baseline. The paper also lacks details on the statistical tests used for significance analysis. Section 4.4 mentions: "Statistical analysis using [Test Method] affirmed the validity of results..." The use of the placeholder '[Test Method]' indicates that the specific statistical tests are not mentioned. The lack of these details makes it difficult to assess the validity of the experimental results and to compare the proposed method with existing approaches. My confidence in this assessment is high, as the paper explicitly omits crucial experimental details.

Finally, the paper's literature review is not comprehensive. While the paper cites relevant works in causal inference, adaptive learning, and NLP, it does not provide a thorough discussion of the existing literature. The paper also does not adequately address the potential limitations of the proposed approach. For example, the paper does not discuss the challenges of identifying causal relationships in observational data or the potential for bias in the adaptive learning process. The lack of a comprehensive literature review and a discussion of limitations weakens the paper's contribution and its overall credibility. My confidence in this assessment is medium, as the paper does cite some relevant works, but it lacks depth and breadth in its literature review and does not address the limitations of the proposed approach.

Suggestions

To address the identified weaknesses, I recommend several concrete improvements. Firstly, the authors must thoroughly revise the paper to correct all grammatical errors, awkward phrasing, and instances of unclear language. This revision should focus on improving the overall clarity and readability of the text. The authors should also ensure that all sentences are grammatically correct and that the language is precise and unambiguous. This is a critical step to ensure that the paper is taken seriously by the academic community. Secondly, the authors need to provide a more detailed description of the

proposed framework. This should include a clear explanation of how the causal features identified by SEMs are mathematically incorporated into the MAB framework. The authors should also provide specific details on the implementation of the causal inference component, including the specific techniques and algorithms used for identifying causal relationships. This should include a detailed explanation of how SEMs and causal diagrams are used to extract causal features. Furthermore, the authors should provide a more detailed explanation of the hybrid MAB model, including the specific implementation of Thompson Sampling and epsilon-greedy strategies. Thirdly, the experimental section needs to be significantly improved. The authors must specify the dataset used for evaluation, including its characteristics and how it was preprocessed. The authors should also provide detailed information about the baseline models used for comparison, including the specific algorithms and configurations used. The authors should also provide details on the statistical tests used for significance analysis, including the specific tests used and the results of these tests. The authors should also consider conducting experiments on multiple datasets to demonstrate the generalizability of the proposed method. Finally, the authors should expand the literature review to include a more comprehensive discussion of the existing literature in causal inference, adaptive learning, and NLP. The authors should also address the potential limitations of the proposed approach, including the challenges of identifying causal relationships in observational data and the potential for bias in the adaptive learning process. This should include a discussion of the assumptions underlying the causal inference methods and the potential impact of these assumptions on the results. By addressing these weaknesses, the authors can significantly improve the quality and credibility of their work.

? Questions

After reviewing the paper, I have several questions that I believe are critical to understanding the proposed framework and its limitations. Firstly, I am curious about the specific techniques used to identify causal relationships from user interaction data. While the paper mentions using SEMs and causal diagrams, it does not provide details on the specific algorithms or methods used for this purpose. For example, how are the causal diagrams constructed? What assumptions are made about the data-generating process? How are latent confounders addressed? Secondly, I would like to understand how the causal features identified by SEMs are mathematically incorporated into the MAB framework. The paper states that these features are integrated, but it does not provide any equations or algorithms to support this claim. How do these causal features influence the reward function or the exploration-exploitation trade-off in the MAB algorithm? Thirdly, I am interested in the specific details of the semantic content mapping component. While the paper mentions using LDA and BERT, it does not provide details on how these techniques are used to align user preferences with recommended items. How are the topics generated by LDA used in the recommendation process? How are the BERT embeddings used to capture semantic similarity between user preferences and items? Fourthly, I am curious about the choice of the specific MAB algorithm. The paper mentions using a hybrid of Thompson Sampling and epsilon-greedy strategies, but it does not provide a rationale for this choice. Why was this specific combination chosen over other MAB algorithms? What are the advantages and disadvantages of this approach? Finally, I would like to know more about the limitations of the proposed approach. What are the potential challenges in applying this framework to real-world datasets? What are the assumptions underlying the causal inference methods, and how might these assumptions impact the results? What are the potential biases that could arise in the adaptive learning process? Addressing these questions would provide a more complete understanding of the proposed framework and its potential limitations.



Scores

Soundness: **1.67**

Presentation: **1.0**

Contribution: **1.33**

Rating: 3.0

Keywords

Extract Keywords

Click the button to extract keywords

Insights

Extract Insights

Click the button to extract insights